INCH-POUND

MIL-M-28787/221A(SH)
AMENDMENT 1
19 November 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, PHASE SHIFT, ANALOG, KEY CODE LEE

This amendment forms a part of Military Specification MIL-M-28787/221A(EC), dated 8 June 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGE 4

TABLE I Delete and substitute new table I as printed on page 2 of this amendment

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4 5 1, fifth line Delete "D through P" and substitute "D through K, M, and P".

4.5.3, third and sixth lines Delete "D through P" and substitute "D through K, M, and P".

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FIGURE 7. Delete and substitute new figure 7 as printed on page 3 of this amendment.

Preparing activity:
Navy - SH

(Project 5963-N781)

Agent: NW

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TABLE I. Module electrical requirements

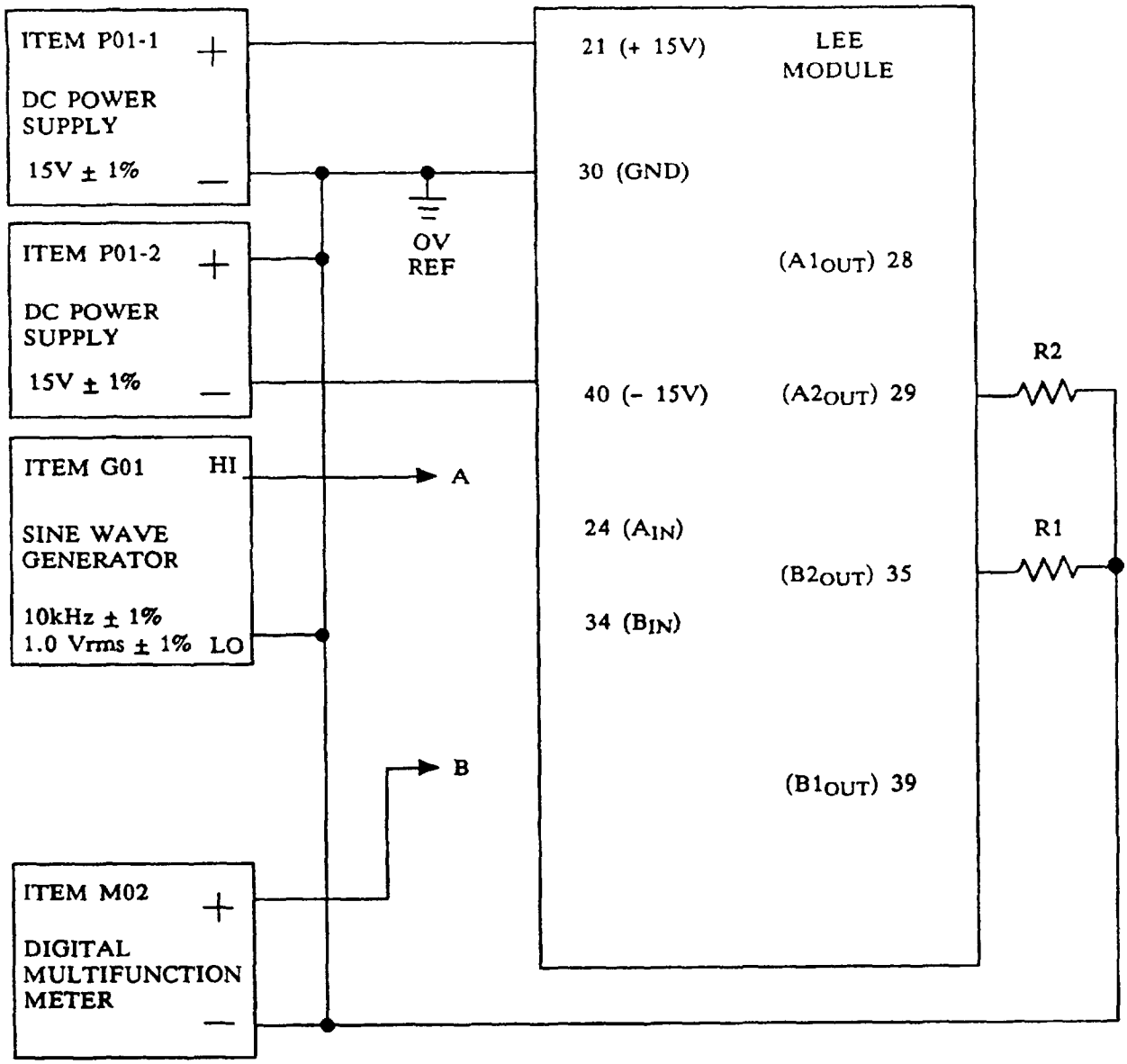
Test Cond.	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0 to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
A2	Isolation, used to unused	II	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1, B2	Output short circuit	III	Q	1	--	1	--	1	--	1	--	Minute
C1, C2	Stress	IV	S	6 72	7 42	6 72	7 42	6 72	7 42	6.72	7 42	V
D1, D2	Resistance check	V	X	2 4	2 6	2 4	2.6	2 4	2 6	2 4	2 6	V
E1-E6	Gain accuracy (3 to 100kHz)	VI	X	0.95	1 05	0 95	1 05	0.95	1.05	0.95	1 05	V
E7, E8	Gain accuracy (100kHz to 1MHz)	VI	S	0 00	1 25	0 00	1 25	0.00	1 25	0 00	1.25	V
F1-F5	Phase accuracy	VII	X	87	93	87	93	85	95	85	95	Degree
G1, G2	Output offset voltage	VIII	S	-25	25	-25	25	-25	30	-25	30	mV
H1, H2	Output noise	IX	S	0	20	0	20	0	30	0	30	µV
J1	Input impedance check	X	S	1.08	1 20	1 08	1.20	1.08	1.20	1.08	1 20	V
J2	Input impedance check	X	S	0 95	1 05	0 95	1 05	0 95	1 05	0 95	1 05	V
K1, K2	Output impedance check	XI	S	14 25	31 5	14 25	31 5	14 25	31 5	14 25	31 5	mV
L1, L2	Quiescent power supply current test	XII	Q	3 5	16 0	3 5	16 0	3	18	3	18	mA
M1, M2	Input common mode voltage swing	XIII	S	6.65	7 35	6.65	7.35	6.65	7 35	6.65	7 35	V
N1-N4	Power supply ripple rejection	XIV	Q	0	4	0	4	0	5	0	5	mV
P1, P2	Power supply transient rise time	XV	S	25	75	25	75	25	75	25	75	µS

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only

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NOTES.
1. R1, R2 = 4.99kΩ ± 1%, 1/8W.
2 Before test condition D1, connect point A to pin 24, point B to pin 28 – adjust G01 to give 5Vrms ± 1% on item M02.
3 Before test condition D2, connect point A to pin 34, point B to pin 39 – adjust G01 to give 5Vrms ± 1% on item M02.

FIGURE 7 Resistance check test circuit for table V.

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